

DC Circuit Experiment Analysis

This infographic presents the basic principles of Direct Current (DC) circuits and demonstrates several simple circuit experiments designed to help understand how electricity behaves in a closed electrical system. The project focuses on the interaction between fundamental electrical components such as a battery (power source), switch, light bulb, motor, wires, and simple conductive materials. By building and observing these circuits, we can better understand how electric current flows, how polarity affects devices, and how different circuit configurations influence the performance of electrical components.

The first part of the project demonstrates the simplest flashlight circuit. In this setup, a battery, switch, and light bulb are connected in a closed loop. When the switch is turned on, the circuit becomes closed, allowing electric current to flow from the positive terminal of the battery, through the switch and the light bulb, and back to the negative terminal. As a result, the light bulb lights up. This experiment helps illustrate the concept of a closed circuit, which is necessary for current to flow.

The second experiment replaces the light bulb with a small propeller motor. In this circuit, electrical energy from the battery is converted into mechanical energy as the motor rotates the propeller. When the wires connected to the battery are reversed, the direction of the current changes, which causes the motor to rotate in the opposite direction. This demonstrates the concept of polarity and shows how the direction of current affects the behavior of certain electrical devices.

The third experiment focuses on a series connection of two light bulbs. In a series circuit, electrical components are connected one after another along the same path. Because the voltage from the battery must be shared between the two bulbs, each bulb receives less voltage than it would individually. As a result, both bulbs glow more dimly. This experiment demonstrates the concept of voltage division and helps explain why devices connected in series may receive less power.

The fourth experiment demonstrates a parallel connection of two light bulbs. In this configuration, both bulbs are connected directly across the battery terminals in separate branches. Each bulb receives the full voltage of the battery, which allows them to shine brightly. However, because more current flows in the circuit, the battery drains more quickly. This experiment helps illustrate the differences between series and parallel circuits, particularly in terms of brightness, current flow, and power consumption.

The final experiment introduces the concept of a homemade fuse using a thin strip of aluminum foil. In this setup, the foil acts as a conductor that melts if too much current flows through the circuit. When excessive current passes through the foil, it heats up and eventually breaks, interrupting the circuit. This simulates how a real electrical fuse works and demonstrates the concept of a short circuit and circuit protection.

In addition to building the circuits, measurements and calculations were performed to better understand the electrical behavior of the system. Using the formulas $P = I \times V$ (power equals current multiplied by voltage) and $R = V / I$ (resistance equals voltage divided by current), different electrical values such as power, current, voltage, and resistance were calculated during the experiment. These calculations help explain how electrical energy is distributed in a circuit and how component limits can be determined.

During the investigation, the voltage supplied to the light bulb was gradually increased. At lower voltage levels, the bulb operated normally. However, when the voltage reached approximately 12.8

volts, the power exceeded about 3 watts, which caused the light bulb to burn out. This observation shows that electrical components have operating limits, and exceeding those limits can damage or destroy them.

Overall, this project helped demonstrate several fundamental electrical concepts, including current flow, circuit polarity, voltage distribution, power consumption, series and parallel connections, and circuit protection mechanisms. It also emphasized the importance of using proper electrical calculations and understanding component limitations to design safe and efficient circuits. Through these experiments, we gained practical experience with basic electrical systems and learned how theoretical formulas apply to real-world circuit behavior.

Circuit	Components	Energy Use	Result
1	Motor + Switch	Low	Simple rotation
2	2 Bulbs + Motor	Medium	Bulbs dimmer
3	Multiple Loads	High	Bright bulbs, battery drains faster